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$$\begin{aligned}& \int \frac{1}{\sqrt{3x^2 + 2}} dx \\& t = \sqrt{3}x \Rightarrow dt = \sqrt{3}dx \Rightarrow dx = \frac{1}{\sqrt{3}}dt \\& = \int \frac{1}{\sqrt{t^2 + 2}} \cdot \frac{1}{\sqrt{3}} dt \\& = \frac{1}{\sqrt{3}} \int \frac{1}{\sqrt{t^2 + 2}} dt \\& = \frac{1}{\sqrt{3}} \ln |t + \sqrt{t^2 + 2}| + C \\& = \frac{1}{\sqrt{3}} \ln |\sqrt{3}x + \sqrt{3x^2 + 2}| + C\end{aligned}$$

References